

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN**

|                                                |                         |                                               |
|------------------------------------------------|-------------------------|-----------------------------------------------|
| <b>Course:</b> Data Handling and Visualization | <b>Code:</b> DSA0601    | <b>CO:</b> CO3                                |
| <b>Duration:</b> 120 min                       | <b>Total Marks:</b> 100 | Reg No : 192424238<br>Name: Jayaprakash Nivas |

**COURSE OUTCOME COVERED**

*CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)*

*Activity Tool : Visualization Critique & Redesign*

*Weightage: 20%*

**Code of Conduct:**

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

| Criteria | Evaluation of Existing Visualization<br>25% | Redesign Strategy<br>25% | Application of Visualization Principles<br>20% | Organization<br>15% | Accuracy<br>10% | Clarity<br>5% | Total |
|----------|---------------------------------------------|--------------------------|------------------------------------------------|---------------------|-----------------|---------------|-------|
| Q1       |                                             |                          |                                                |                     |                 |               |       |
| Q2       |                                             |                          |                                                |                     |                 |               |       |
| Q3       |                                             |                          |                                                |                     |                 |               |       |
| Q4       |                                             |                          |                                                |                     |                 |               |       |
| Q5       |                                             |                          |                                                |                     |                 |               |       |

# AT3 – Visualization Critique & Redesign

## *R Programming Assessment – Distribution and Proportion Visualization*

Instructions: For each question, use R/RStudio to reproduce the given visualization, critique its effectiveness, identify misleading or inappropriate visual encodings, and redesign it using a more suitable visualization. Include the R code, redesigned visualization, and a concise justification of your design decisions.

### Question 1 – Critique and Redesign an ECDF and Q-Q Plot

#### **Dataset:**

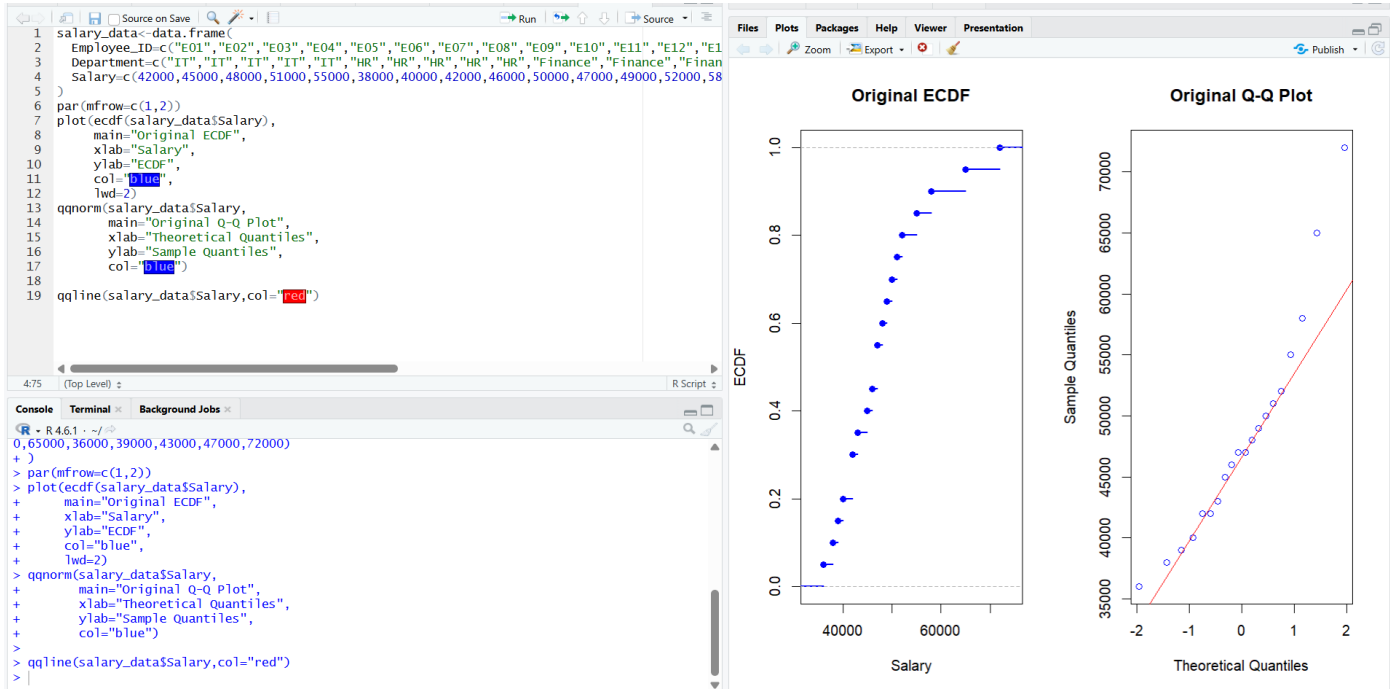
Salary\_Distribution.csv Dataset:

| Employee_ID | Department | Salary |
|-------------|------------|--------|
| E01         | IT         | 42000  |
| E02         | IT         | 45000  |
| E03         | IT         | 48000  |
| E04         | IT         | 51000  |
| E05         | IT         | 55000  |
| E06         | HR         | 38000  |
| E07         | HR         | 40000  |
| E08         | HR         | 42000  |
| E09         | HR         | 46000  |
| E10         | HR         | 50000  |
| E11         | Finance    | 47000  |
| E12         | Finance    | 49000  |
| E13         | Finance    | 52000  |
| E14         | Finance    | 58000  |
| E15         | Finance    | 65000  |
| E16         | Marketing  | 36000  |
| E17         | Marketing  | 39000  |
| E18         | Marketing  | 43000  |
| E19         | Marketing  | 47000  |
| E20         | Marketing  | 72000  |

#### **Assessment Task**

A salary analyst has created an ECDF and a Q-Q plot to study employee salary distribution, but the charts use unclear labels, inappropriate scaling, and excessive decoration.

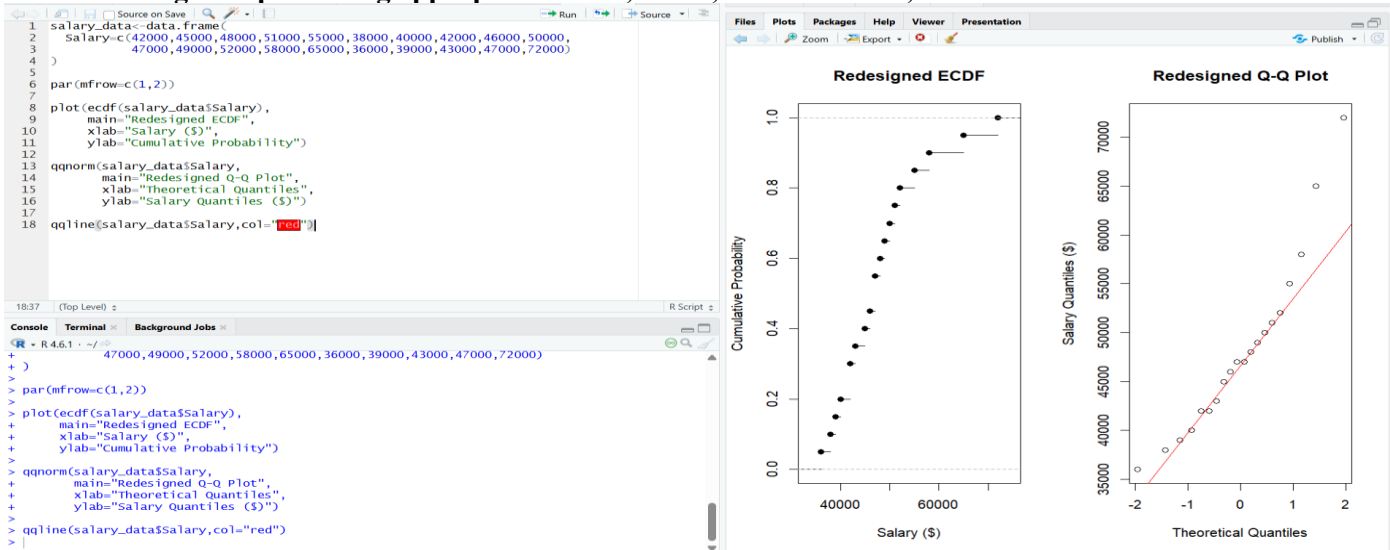
## 1. Create the two original-style plots in R.



## 2. Identify at least three visualization problems.

The visualization has unclear labels, inappropriate scaling, and excessive decoration. These issues make the salary distribution difficult to interpret and reduce the clarity of the ECDF and Q-Q plots.

## 3. Redesign the plots using appropriate scales, labels, reference lines, and minimal visual clutter.



## 4. Explain how the redesigned plots improve interpretation of skewness, outliers, and normality.

The redesigned ECDF makes the salary distribution easier to understand through clear labels and an appropriate scale. The Q-Q plot uses a reference line, making deviations from normality easier to identify. The simple design also makes skewness and outliers easier to observe.

## 5. State whether a Q-Q plot alone is sufficient to conclude that the data are normally distributed.

No. A Q-Q plot alone is not sufficient to conclude that the data are normally distributed. It should be considered together with the data distribution and other appropriate normality checks.

# Question 2 – Critique and Redesign Many Distributions Using Bean Plots

Dataset: Student\_Scores.csv

Dataset:

| Student_ID | Course     | Score |
|------------|------------|-------|
| S01        | Python     | 62    |
| S02        | Python     | 68    |
| S03        | Python     | 74    |
| S04        | Python     | 81    |
| S05        | Python     | 88    |
| S06        | Statistics | 58    |
| S07        | Statistics | 64    |
| S08        | Statistics | 71    |
| S09        | Statistics | 79    |
| S10        | Statistics | 86    |
| S11        | DataViz    | 55    |
| S12        | DataViz    | 61    |
| S13        | DataViz    | 69    |
| S14        | DataViz    | 77    |
| S15        | DataViz    | 84    |
| S16        | AI         | 65    |
| S17        | AI         | 72    |
| S18        | AI         | 78    |
| S19        | AI         | 87    |
| S20        | AI         | 93    |

## Assessment Task

A college report uses four separate histograms with different bin widths to compare course scores. The comparison is difficult because the scales and layouts are inconsistent.

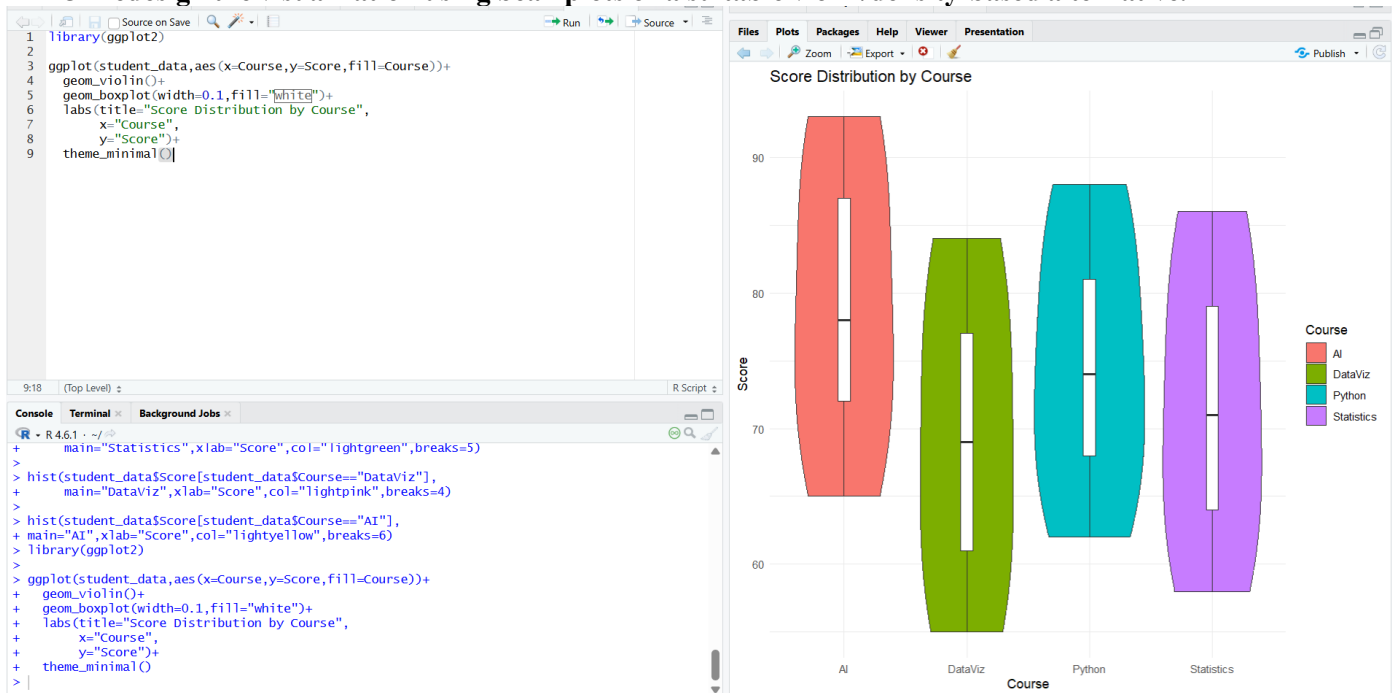
### 1. Reproduce the problematic visualization using R.



## 2. Critique the use of separate histograms, inconsistent scales, and bin widths.

The separate histograms are difficult to compare because they use different bin widths, inconsistent scales, and different layouts. These differences can make the distributions appear more or less variable even when the underlying data are similar. A common scale and consistent bin width would make comparison easier and more accurate.

## 3. Redesign the visualization using bean plots or a suitable violin/density-based alternative.



## 4. Add an appropriate common scale and clear course labels.

Use the same score scale for all four courses and provide clear course labels. A common scale allows the distributions to be compared fairly and makes differences in scores easier to identify.

## 5. Explain which course has the highest concentration of scores and which course shows the greatest spread.

- AI has the highest concentration of scores toward the upper score range, with scores ranging from 65 to 93.
- DataViz shows the greatest spread, with scores ranging from 55 to 84, giving a range of 29 points.

## Question 3 – Critique and Redesign Proportion Visualizations

### Dataset:

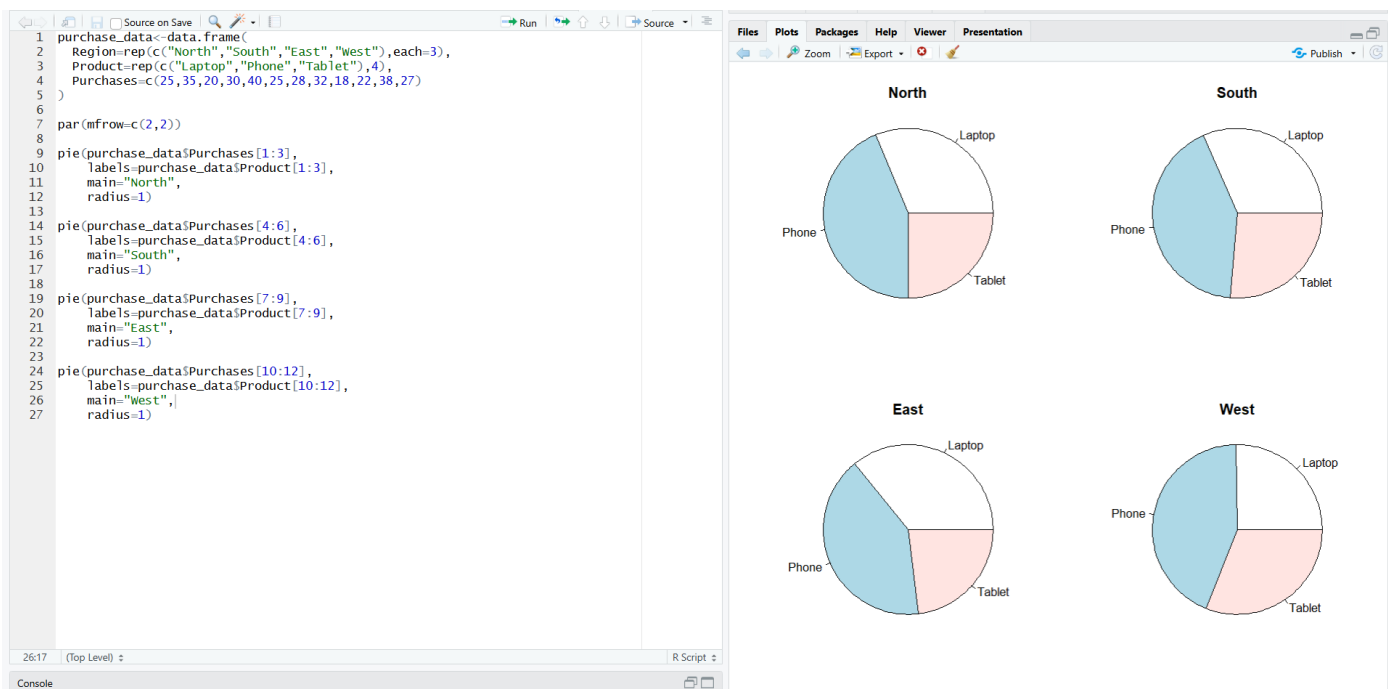
Regional\_Purchases.csv Dataset:

| Region | Product | Purchases |
|--------|---------|-----------|
| North  | Laptop  | 25        |
| North  | Phone   | 35        |
| North  | Tablet  | 20        |
| South  | Laptop  | 30        |
| South  | Phone   | 40        |
| South  | Tablet  | 25        |
| East   | Laptop  | 28        |
| East   | Phone   | 32        |
| East   | Tablet  | 18        |
| West   | Laptop  | 22        |
| West   | Phone   | 38        |
| West   | Tablet  | 27        |

### Assessment Task

A manager presents product proportions using four pie charts, one for each region. The charts use different starting angles, 3-D effects, and inconsistent labeling.

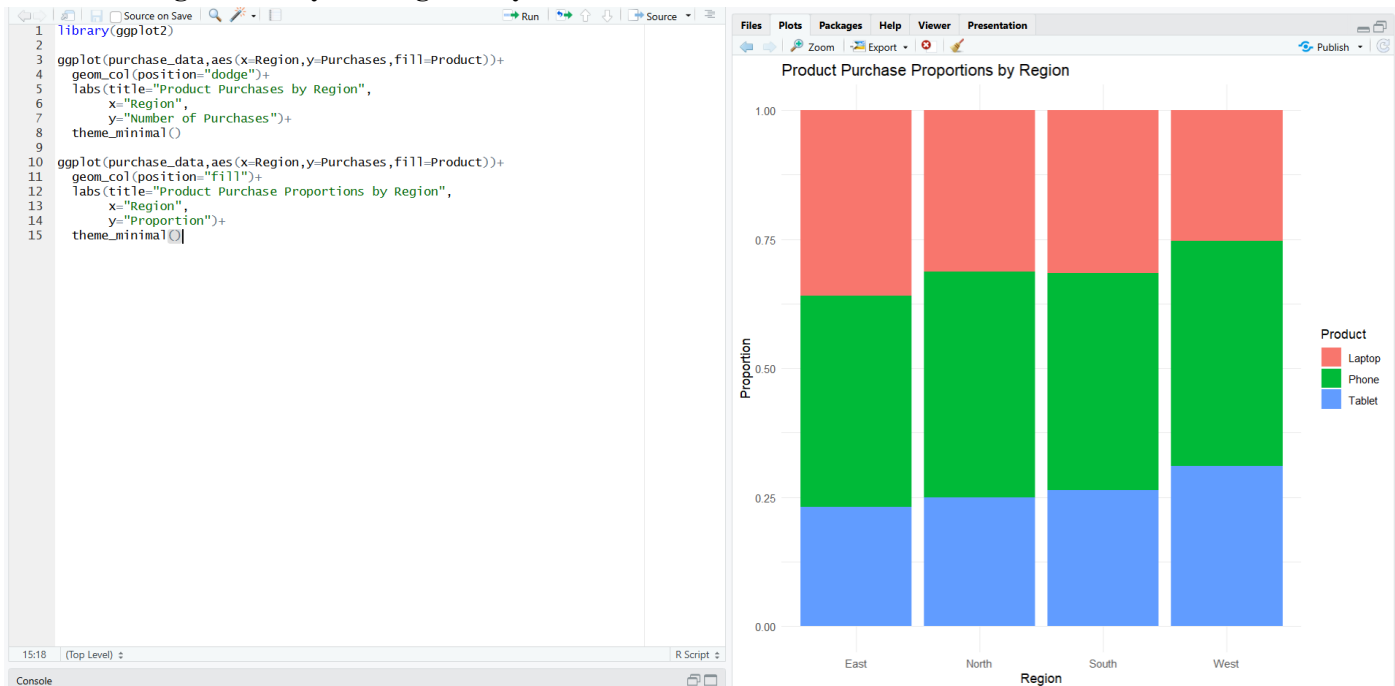
#### 1. Create a representative version of the misleading visualization.



#### 2. Identify at least four design problems.

- Different starting angles make the pie charts difficult to compare.
- 3-D effects can distort the perceived size of the slices.
- Inconsistent labeling reduces clarity.
- Comparing proportions across four separate pie charts is difficult.
- The visual design can make small differences appear larger or smaller than they actually are.

### 3. Redesign the analysis using side-by-side bars and a 100% stacked bar chart.



#### 4. Compare the two redesigns and determine which is better for comparing regional proportions.

The 100% stacked bar chart is better for comparing regional proportions because every region has the same total height of 100%. This makes it easier to compare the relative proportions of Laptop, Phone, and Tablet purchases across regions.

The side-by-side bar chart is better for comparing the actual number of purchases for each product.

#### 5. Explain why 3-D effects and inconsistent scales can lead to misleading conclusions.

3-D effects can distort the perceived size and angle of pie-chart sections, making some values appear larger or smaller than they actually are. Inconsistent scales or labeling can also exaggerate or hide differences between regions. Therefore, simple 2-D visualizations with consistent scales provide more accurate comparisons.

## Question 4 – Critique and Redesign Nested Proportions

### Dataset:

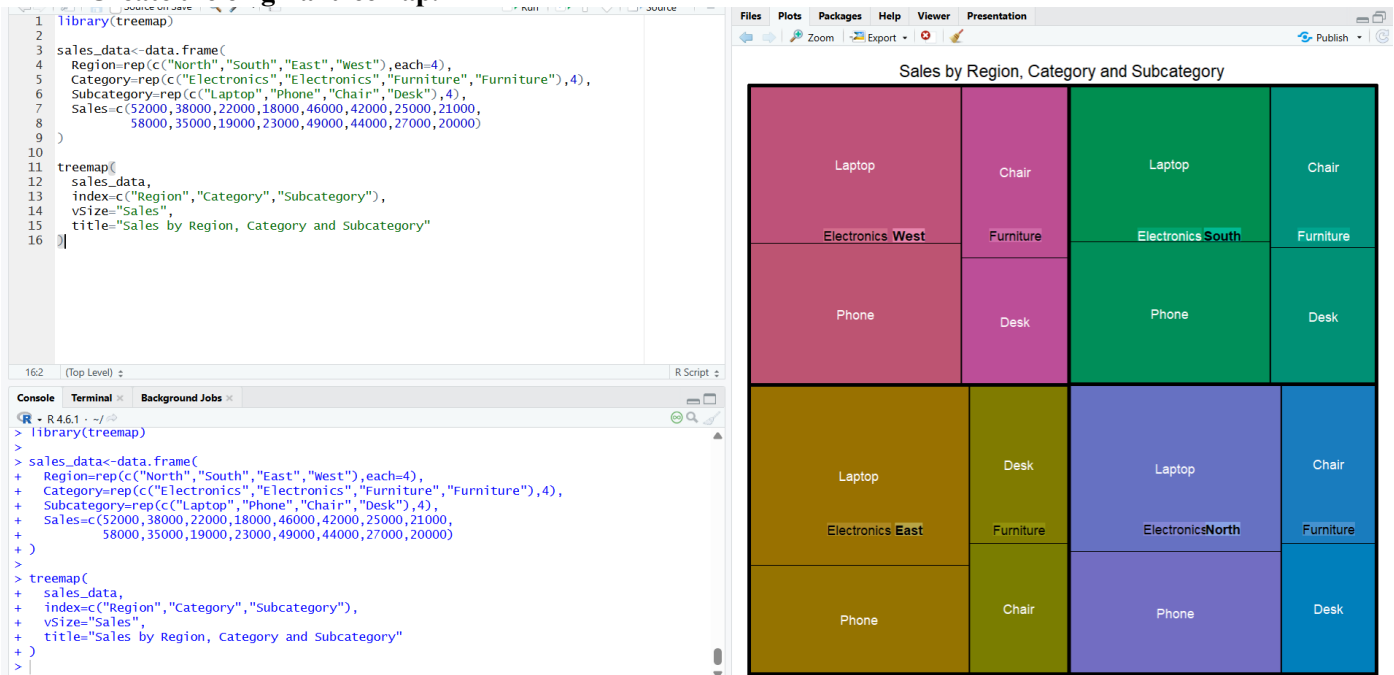
Sales\_Hierarchy.csv Dataset:

| Region | Category    | Subcategory | Sales |
|--------|-------------|-------------|-------|
| North  | Electronics | Laptop      | 52000 |
| North  | Electronics | Phone       | 38000 |
| North  | Furniture   | Chair       | 22000 |
| North  | Furniture   | Desk        | 18000 |
| South  | Electronics | Laptop      | 46000 |
| South  | Electronics | Phone       | 42000 |
| South  | Furniture   | Chair       | 25000 |
| South  | Furniture   | Desk        | 21000 |
| East   | Electronics | Laptop      | 58000 |
| East   | Electronics | Phone       | 35000 |
| East   | Furniture   | Chair       | 19000 |
| East   | Furniture   | Desk        | 23000 |
| West   | Electronics | Laptop      | 49000 |
| West   | Electronics | Phone       | 44000 |
| West   | Furniture   | Chair       | 27000 |
| West   | Furniture   | Desk        | 20000 |

### Assessment Task

An organization uses a treemap to communicate Region → Category → Subcategory sales. Small areas have unreadable labels and the color scale does not clearly communicate the hierarchy.

#### 1. Create the original treemap.

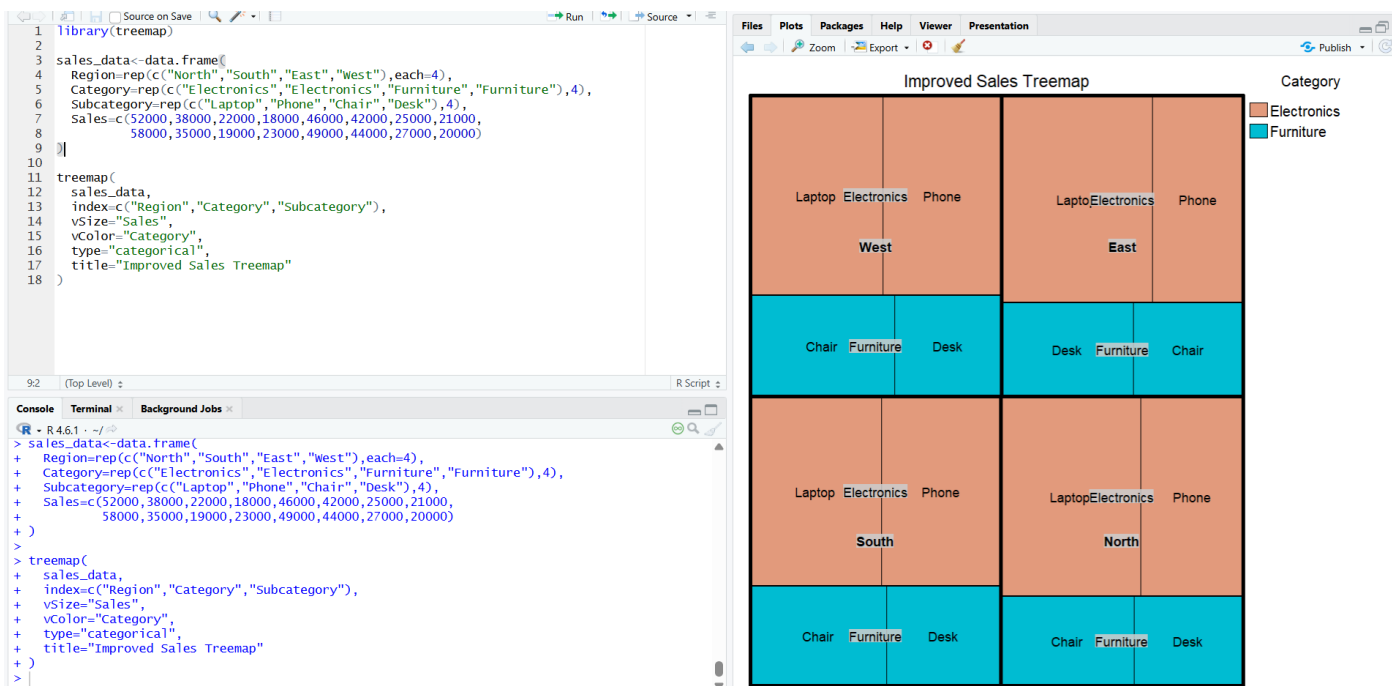


#### 2. Critique the size, labeling, color, and hierarchy encodings.

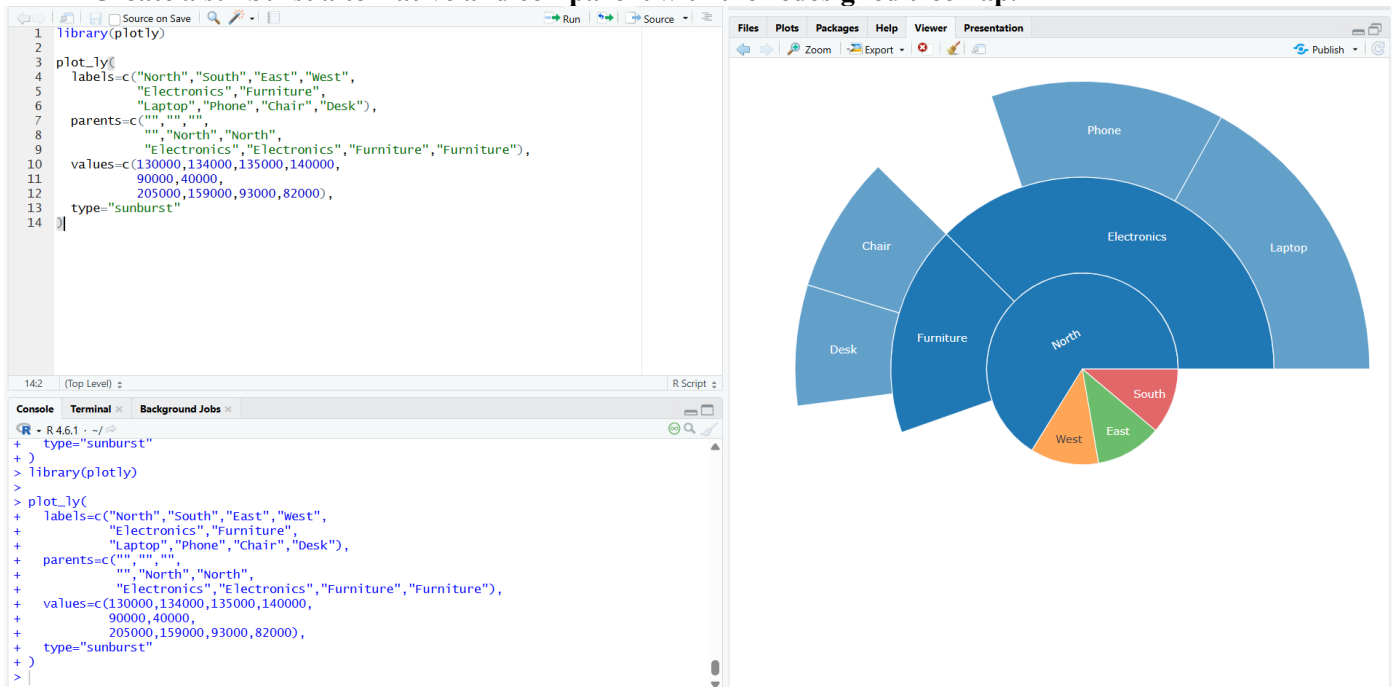
The treemap has several visualization problems. Small areas can have unreadable labels, making some subcategories difficult to identify. The color scale does not clearly communicate the hierarchy between Region, Category, and Subcategory. Similar colors can also make different categories difficult to distinguish. Overall, the hierarchy is not communicated clearly enough.

#### 3. Redesign the visualization as an improved treemap.





#### 4. Create a sunburst alternative and compare it with the redesigned treemap.



#### 5. Recommend one visualization for a management presentation and justify your choice.

I recommend the improved treemap for a management presentation because it provides a clear and compact view of sales values. It makes it easier to identify the largest and smallest contributors while still showing the Region ->Category-> Subcategory hierarchy.

## Question 5 – Critique and Redesign Flow-Based Proportions

### Dataset:

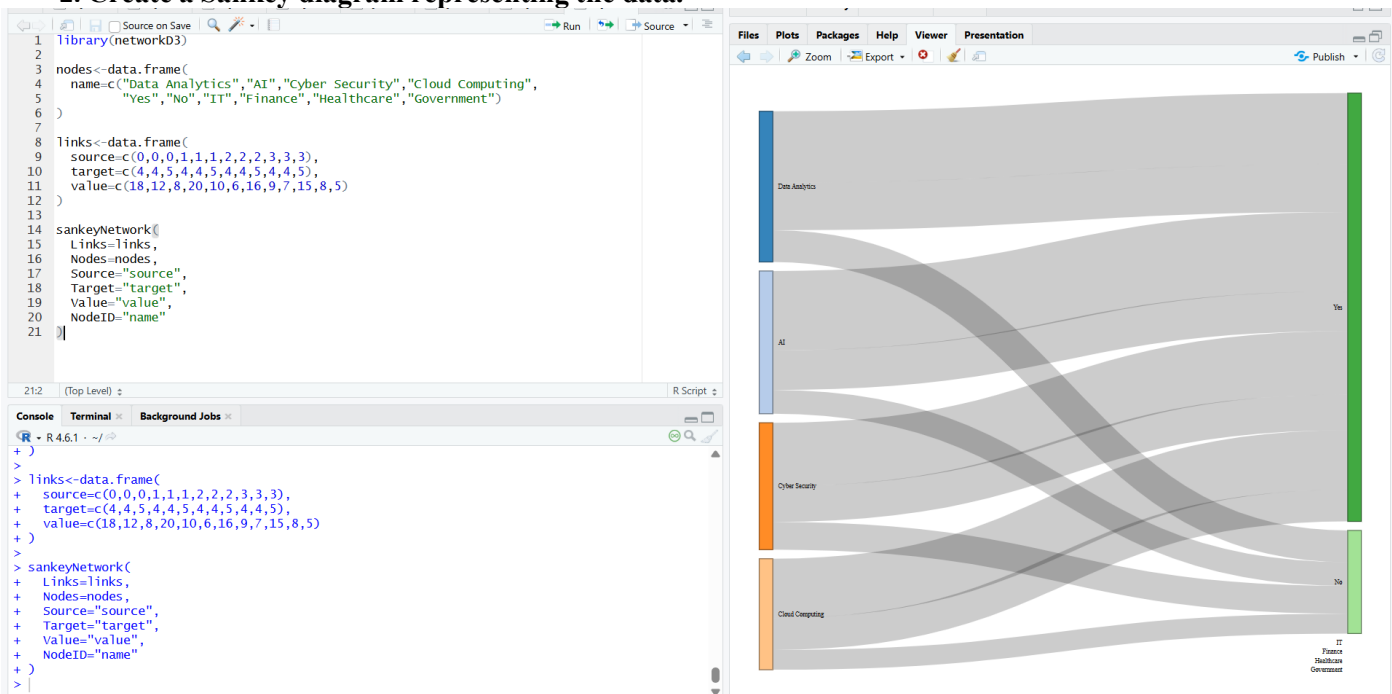
Career\_Flow.csv Dataset:

| Specialization  | Internship | Placement_Sector | Students |
|-----------------|------------|------------------|----------|
| Data Analytics  | Yes        | IT               | 18       |
| Data Analytics  | Yes        | Finance          | 12       |
| Data Analytics  | No         | IT               | 8        |
| AI              | Yes        | IT               | 20       |
| AI              | Yes        | Healthcare       | 10       |
| AI              | No         | IT               | 6        |
| Cyber Security  | Yes        | IT               | 16       |
| Cyber Security  | Yes        | Finance          | 9        |
| Cyber Security  | No         | Government       | 7        |
| Cloud Computing | Yes        | IT               | 15       |
| Cloud Computing | Yes        | Finance          | 8        |
| Cloud Computing | No         | Government       | 5        |

### Assessment Task

A placement office uses a Sankey diagram to show Specialization → Internship → Placement Sector. The original chart uses excessively similar colors, unlabelled flows, and proportions that are difficult to compare.

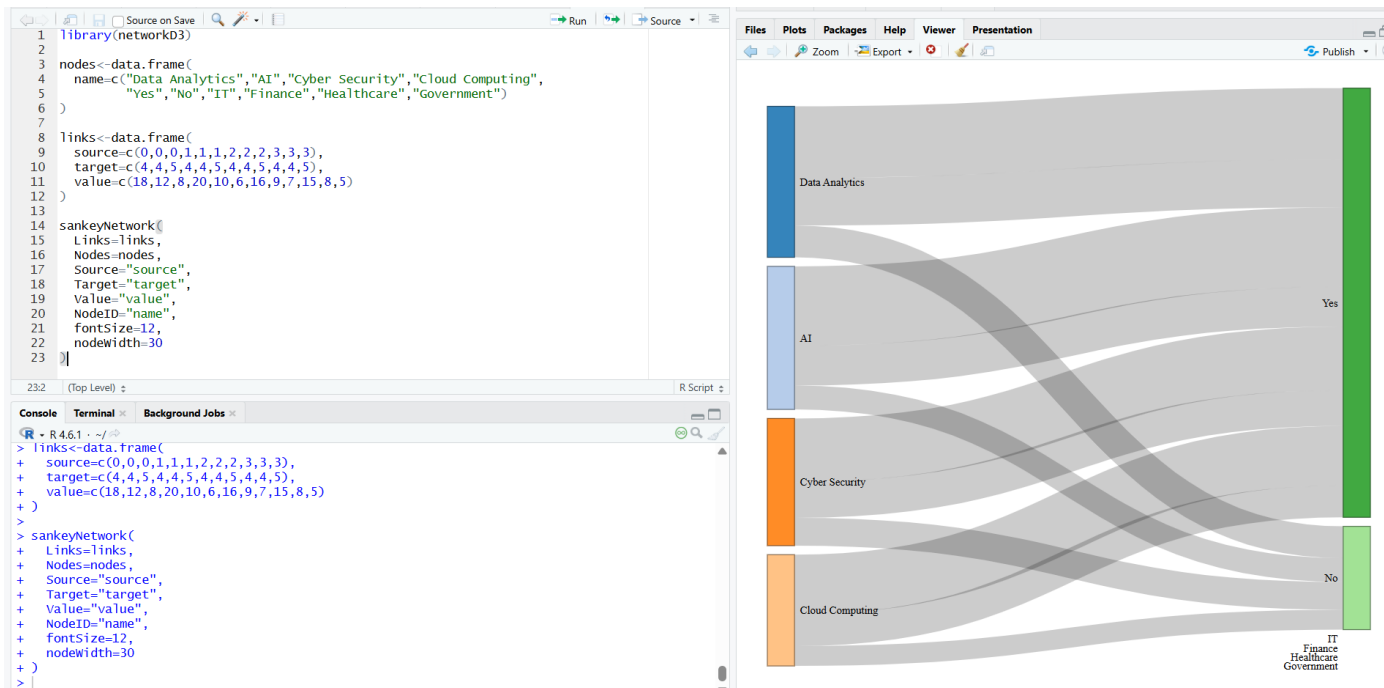
#### 1. Create a Sankey diagram representing the data.



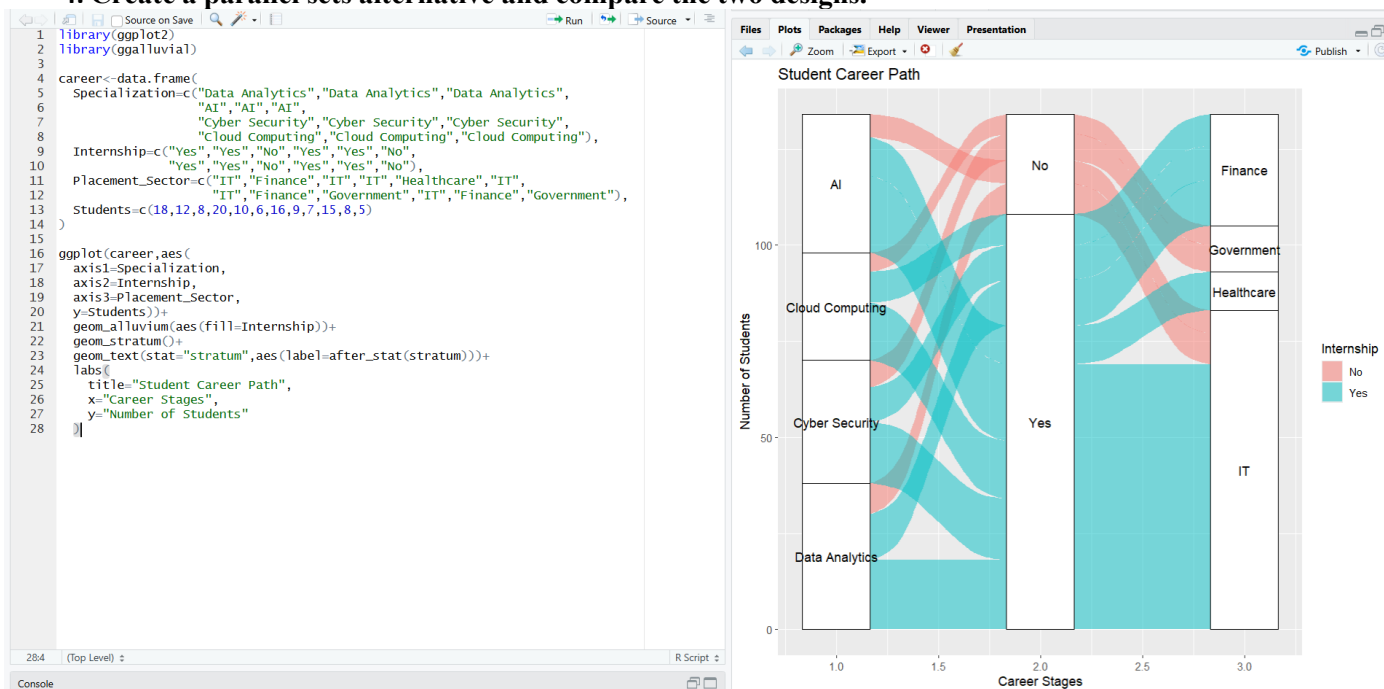
#### 2. Identify at least four potential sources of confusion or misleading interpretation.

- Excessively similar colors make different flows difficult to distinguish.
- Unlabelled flows make the meaning of the connections unclear.
- Difficult-to-compare proportions make major and minor flows harder to identify.
- Poor visual hierarchy makes it difficult to follow the flow from Specialization → Internship → Placement Sector.

### 3. Redesign the Sankey diagram with meaningful labels, consistent flow representation, and an appropriate visual hierarchy.



### 4. Create a parallel sets alternative and compare the two designs.



### 5. Explain which visualization better communicates the major student flows and why.

The Sankey diagram better communicates the major student flows because the width of each flow can represent the number of students. This makes the larger flows easier to identify and understand. The parallel sets visualization is useful for showing relationships between the categorical variables, but Sankey is more effective for communicating flow magnitude.

## AT3 Expected Student Output

- R program/script used to create and redesign the visualization.
- Original/problematic visualization reproduced or clearly simulated.
- Written critique identifying visualization and design problems.
- Redesigned visualization with improved visual encodings.
- Brief justification explaining why the redesign is more accurate, readable, and less misleading.
- Final recommendation where comparison of alternative visualizations is requested.

## Suggested AT3 Assessment Rubric

| Criterion                                  | Weight |
|--------------------------------------------|--------|
| Identification of visualization problems   | 20%    |
| Correct R programming and data preparation | 20%    |
| Quality and appropriateness of redesign    | 30%    |
| Justification and visualization critique   | 20%    |
| Clarity, labeling, and presentation        | 10%    |